⌈Def.
Let G be a group and A be a non-empty subset. A map

$$\alpha : G \times A \to A : (g, a) \mapsto g.a$$

is called a group action of G on A if it satisfies
1) For all $a \in A$, $1_G \cdot A = A$
2) For all $g_1, g_2 \in A$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

⌈Prop. Given a group action α of G on A, there exists a homomorphism

$$\varphi_\alpha : G \to S_A : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : A \to A : a \mapsto ga.$$

which is called the permutation representation associated to the given action.
Conversely, given a homomorphism $\varphi : G \to S_A$, there exists a group action

$$\alpha_\varphi : G \times A \to A : (g, a) \mapsto \varphi(g)(a)$$

Proof. We only need to check that the given map is well-defined,
Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$, therefore $\sigma_g \in S_A$.

「Thm.

Let G be a group and $H \leq G$ be a subgroup.

Грummit. Sec 4.2.